MANYARA REGION
FORM FOUR MOCK EXAMINATION 2021
032/2A **CHEMISTRY 2A**
ACTUAL PRACTICAL A
MARKING SCHEME

1. Table of results

Titration number	Pilot	1	2	3
Final burette reading (cm^3)	20.50	40.50	20.00	40.00
Initial burette reading(cm^3)	0.00	20.50	0.00	20.00
Volume used(cm^3)	**20.50**	**20.00**	**20.00**	**20.00**

(04 marks)

Pipette used was 20cm^3

Note: if 25.00cm^3 pipette was used the volume of acid will be 25.00cm^3

Average volume = $\frac{V_1 + V_2 + V_3}{3}$ **(00$\frac{1}{2}$ marks)**

= $\frac{20.00cm^3 + 20.00cm^3 + 20.00cm^3}{3}$ **(00$\frac{1}{2}$ marks)**

= $\frac{60}{3}$ **(00$\frac{1}{2}$ marks)**

= 20.00cm^3

Therefore the mean titre volume is = 20.00cm^3 **(01marks)**

(a) (i) $\mathbf{20.00cm^3}$of solution **P** required $\mathbf{20.00cm^3}$ of solution **Q** for complete reaction **(01 mark)**

(ii) A balanced chemical equation for the reaction between solution **P** and **Q**.

$\mathbf{XOH_{(aq)} + HCl \rightarrow XCl_{(aq)} + H_2O_{(l)}}$ **(02 marks)**

(iii) The colour change during titration was from **yellow** to **pink**. **(01 mark)**

(b) Calculate the:-

(i) Molarity of solution **Q (HCl acid)**

Data given

Concentration =3.65g/dm^3

Molar mass of HCl =36.5g/mol **(00$\frac{1}{2}$ marks)**

Molarity of HCl =?

From

Molarity = $\frac{\text{concentration}}{\text{Molar mass}}$ **(00$\frac{1}{2}$ marks)**

= $\frac{3.65}{36.5}$ **(00$\frac{1}{2}$ marks)**

= 0.1M

Therefore molarity of HCl is 0.1M **(01mark)**

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(ii) Find molarity of **P (XOH)**

Data given

Concentration = 4g/dm^3

Ma =0.1

Va = 20cc

Vb = 20cc **(00$\frac{1}{2}$ marks)**

Mb = ?

na =1

nb = 1

From

$$\frac{MaVa}{MbVb} \times \frac{na}{nb} \quad \textbf{(00}\tfrac{1}{2}\textbf{ marks)}$$

$$MbVbna = MaVanb$$

$$\frac{MbVbna}{Vbna} = \frac{MaVanb}{Vbna} \quad \textbf{(00}\tfrac{1}{2}\textbf{ marks)}$$

$$Mb = \frac{MaVanb}{Vbna} \quad \textbf{(00}\tfrac{1}{2}\textbf{ marks)}$$

$$Mb = \frac{0.1x\ 20\ x\ 1}{20\ x\ 1} \quad \textbf{(00}\tfrac{1}{2}\textbf{ marks)}$$

= **0.1M**

Therefore the molarity of XOH is 0.1M (01mark)

(iii) Find molar mass of **P (XOH)**

Data given

Concentration =4g/dm^3

Molar mass of XOH = ?

Molarity of XOH =0.1M **(00$\frac{1}{2}$ mark)**

From

$$\text{Molarity} = \frac{\text{concentration}}{\text{Molar mass}} \quad \textbf{(00}\tfrac{1}{2}\textbf{ mark)}$$

$$1$$

Molarity x molar mass = concentration

$$\frac{\text{Molarity x molar mass}}{\text{Molarity}} = \frac{\text{concentration}}{\text{molarity}} \quad \textbf{(00}\tfrac{1}{2}\textbf{ mark)}$$

Molar mass = concentration

Molarity **(00 $\frac{1}{2}$ mark)**

= 4 **(00 $\frac{1}{2}$ mark)**

0.1

= 40

Therefore the molar mass of XOH is 40g/mol **(01mark)**

(c) (i) Atomic mass of X.

Molecular formula = molar mass **(00 $\frac{1}{2}$ mark)**

XOH= 40 **(00 $\frac{1}{2}$ mark)**

x + (16x1) +1 = 40

x +16 +1 =40 **(00 $\frac{1}{2}$ mark)**

x +17 =40 **(00 $\frac{1}{2}$ mark)**

x + 17 -17 = 40 -17 **(00 $\frac{1}{2}$ mark)**

x = 23

Therefore; the atomic mass of x is 23. **(00 $\frac{1}{2}$ mark)**

(ii) The element x in XOH is **sodium.(02 marks)**

2. (a)

S/N	Observation	Inference
(a)	Light green crystalline salt was observed **(01 mark)**	Fe^{2+}, Fe^{3+},Cu^{2+} may be present **(01 mark)**
(b)	soluble forming pale green solution was observed **(01 mark)**	Fe^{2+} may be present **(01 mark)**
(c)	No gas evolved **(01 mark)**	SO_4^{2-} may be present **(01 mark)**
(d)	-Reddish brown residue observed **(01 mark)**	Fe^{2+}, Fe^{3+} may be present **(01 mark)**
	Colourless gas which turn filter paper dipped in acidified potassium dichromate solution from yellow to green was observed **(01 mark)**	SO_4^{2-} of Fe^{2+} , Cu^{2+} may be present **(01 mark)**

(e)	soluble forming pale green solution was observed **(01 mark)**	Fe^{2+} may be present **(01 mark)**
	(i) Green precipitate was formed, insoluble in excess which turned brown on standing **(01 mark)**	Fe^{2+} may be present **(01 mark)**
	(ii) White precipitate was observed **(01 mark)**	SO_4^{2-} present and confirmed **(01 mark)**
	(i) Deep blue precipitate was observed **(01 mark)**	Fe^{2+} present and confirmed **(01 mark)**

(i) The cation in sample **R** is Fe^{2+} **(01 mark)**

(ii) Anion in Sample **R** is SO_4^{2-} **(01 mark)**

(iii) The name of sample **R** is Iron (II) sulphate **(01 mark)**

(iv) The chemical formula of sample **R** is $FeSO_4$ **(02 marks)**

(b) When Barium chloride was added to the sample solution white precipitate of barium sulphate was formed. **(02 marks)**